Proceedings of the 6th International Conference on Pervasive Computing and Social Networking (ICPCSN-2026)
DVD Part Number: CFP26EZ5-DVD; ISBN: 979-8-3315-7235-8

Privacy-Preserving Personalized Federated Learning for Heterogeneous Intrusion Detection Systems

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Abstract—The presented paper proposes a Privacy-Preserving Personalized Federated Learning (PP-PFL) framework for Network Intrusion Detection Systems (IDS). This framework addresses three major challenges: cross-domain data heterogeneity, worst-case client fairness, and privacy leakage in federated environments. Traditional FedAvg suffers severe degradation under non-IID data, where worst-case client accuracy can drop to 18.41%. For the most disadvantaged clients, our approach bridges this gap. By combining local encoders and a shared alignment layer with differential privacy, we've created a system that balances security with performance through synergistic optimization using Focal Loss with Cosine Annealing learning rate scheduling. Focal Loss alone degrades worst-case performance due to optimization instability.Experimental evaluation demonstrates 91.24% NSL-KDD accuracy, 58.45% worst-case client improvement (+40.04%), and 47% variance reduction. Communication overhead remains constant (~70KB/client) and inference latency remains below 0.01 ms, demonstrating suitability for practical deployment.

***Index Terms*—Federated learning, intrusion detection systems, personalized federated learning, differential privacy, non-IID data, synergistic optimization**

I. INTRODUCTION

Network Intrusion Detection Systems (IDS) are essential for protecting modern networks from increasingly sophisticated cyber threats. Traditional intrusion detection systems use centralized machine learning, where all data is shared with a server that aggregates it. This has severe limitations, like privacy concerns and data heterogeneity. To tackle this, Federated Learning has been introduced, where the model is trained without sharing raw data. This approach fails under non-IID conditions, as it assumes all data is homogeneous in nature. In this work, we propose a personalized federated learning framework tailored for intrusion detection under extreme data heterogeneity. The framework employs client-specific local encoders to capture environment-dependent features while sharing higher-level representations through a global classifier. It enables collaboration without compromising privacy. Rather than focusing solely on aggregate accuracy, our study emphasizes client-level robustness, fairness, and stability under non-IID conditions. However, deploying federated intrusion detection systems in real-world environments introduces additional challenges including communication latency, heterogeneous client resources, and synchronization delays that may impact real-time intrusion detection performance.

The main contributions of this paper are summarized as follows:
1. We propose a personalized federated learning architecture for intrusion detection that balances local adaptation with global knowledge sharing under heterogeneous data distributions.
2. We systematically analyze the interaction between optimization components in personalized federated learning, highlighting the importance of coordinated design choices for stable and fair training.
3. We provide a comprehensive experimental evaluation on benchmark intrusion detection datasets, focusing on worst-case client performance, variance across clients, and convergence behavior as key indicators of practical deployability.

II. RELATED WORK

Federated Learning (FL) was introduced as a decentralized learning paradigm that enables collaborative training for multiple clients without ever sharing raw data, thereby addressing privacy and data ownership concerns in distributed environments [1]. McMahan et al. proposed the FedAvg algorithm that aggregates locally trained model updates at a central server, but it assumes that all client data distributions are similar, which is not always the case in real-world deployments [1]. Studies have shown the limitations of federated learning under non-IID data environments. Li et al. highlighted that heterogeneity across clients degrades performance, makes training unstable, and results in very slow convergence [2]. Konečný et al. further explored federated optimization strategies beyond traditional datacenter environments, addressing the unique challenges posed by distributed data across heterogeneous clients [3].

Privacy-preserving mechanisms are a critical component of federated learning. Individual client updates should be securely aggregated so that the server cannot inspect them, thereby guaranteeing privacy in practical deployments [4]. Differential privacy (DP) further limits information leakage by injecting calibrated noise into the model updates [5]. Prior work has also demonstrated how effective differential privacy is in protecting sensitive information during collaborative learning, as well as how it introduces a trade-off between privacy and model utility [5], [6].

In the domain of intrusion detection systems (IDS), machine learning-based approaches have been extensively studied using benchmark datasets such as NSL-KDD and CICIDS2017 [7], [8]. These datasets capture diverse attack patterns and network behaviors, making them suitable for evaluating IDS models.Recent comprehensive surveys analyze federated learning systems and open challenges [13], [14], highlighting persistent issues in optimization under heterogeneity [15]. Focal Loss has been successfully applied to address class imbalance in computer vision and classification tasks [9], [10], while learning rate scheduling techniques such as cosine annealing and cyclical learning rates [11], [12] have demonstrated effectiveness in improving convergence stability and training dynamics.

III. PROPOSED METHODOLOGY

The IDS framework is based on a personalized federated learning architecture that handles heterogeneous data while ensuring privacy. It contains multiple clients that share model components with a central server, which aggregates them. Each client has a local encoder that

trains on its own data without ever sharing it with the server. This encoder learns environment-specific patterns from network traffic, which helps personalize the model and reduces non-IID data problems.

The alignment layer maps encoders' latent representations to a shared feature space, making server aggregation effective despite data heterogeneity. Each round, clients perform local training on their private datasets and send computed updates to the central server for aggregation. The updated model is then broadcast back to clients for the next training round. Before aggregation, we apply differential privacy to enhance privacy. We implement gradient clipping and Gaussian noise injection to prevent reverse-engineering of raw data from model updates..

A.Mathematical Formulation

Let $C = \{c_1, c_2, \dots, c_k\}$ represent K federated clients where each client holds private dataset D_i.
Each client model is defined as:
$f_i(x) = g(h(e_i(x)))$
where:
$e_i \rightarrow$ client specific encoder
$h \rightarrow$ shared alignment layer
$g \rightarrow$ global classifier

Local training objective:
$L_i = L_FL(f_i; D_i) + (\lambda/2)||\theta_shared_i - \theta_global||^2$
where λ controls personalization strength.

Focal Loss:
$L_FL(p_t) = -\alpha_t(1-p_t)^\gamma \log(p_t)$
where:
$\gamma = 2$
α_t = class weight

Federated aggregation:
$\theta_global(t+1) = (1/K) \Sigma\, \theta_shared_i(t)$

Differential privacy update:
$\theta_DP = clip(\theta, C) + N(0, \sigma^2C^2I)$
Noise multiplier:
$\sigma = \sqrt{(2 \ln(1.25/\delta))} / \varepsilon$
Cosine Annealing learning rate:
$\eta_t = \eta_min + ½(\eta_max - \eta_min)(1 + \cos(\pi t/T))$
where:
$\eta_max = 1\times10^{-3}$
$\eta_min = 1\times10^{-5}$
T = total rounds

IV. SYSTEM ARCHITECTURE

We designed a hybrid architecture that combines personalized and federated learning. Each client runs a local encoder that captures dataset-specific characteristics. Raw network features are transformed into latent representations, which the alignment layer maps to a common feature space. The global classifier then assigns them to different intrusion classes.

Raw data never reaches the central server—only model updates are communicated. This design minimizes communication costs and prevents exposure of sensitive network traffic. The architecture handles both IID and non-IID data distributions, making it suitable for real-world deployments.

Since NSL-KDD and CICIDS2017 produce different latent distributions, direct classifier sharing would lead to representation mismatch. The alignment layer transforms client embeddings into a normalized common feature space

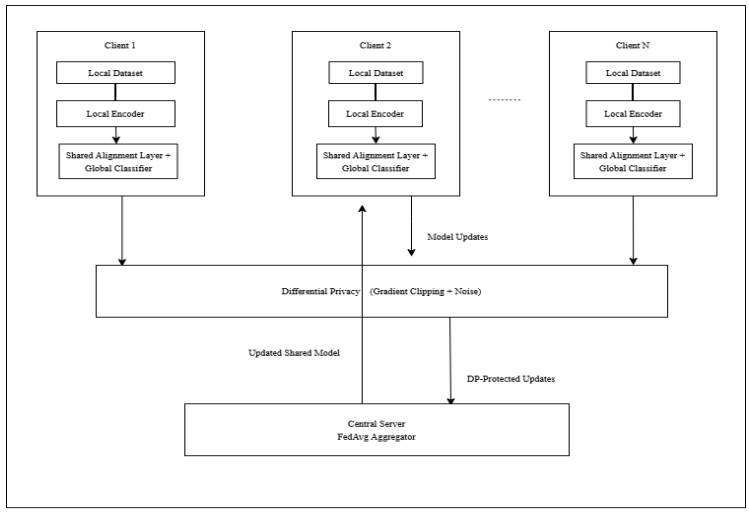

Fig. 1. System architecture

V. EXPERIMENTAL SETUP

A. Datasets

The framework is evaluated on two datasets, NSL-KDD and CICIDS2017. Both are publicly available and contain various network traffic characteristics that allow us to evaluate heterogeneous data distributions. NSL-KDD is a traditional dataset that has older attack patterns. CICIDS2017 is a modern dataset that contains contemporary attack patterns generated from realistic traffic traces.

The datasets help the experiments to reflect realistic cross-domain heterogeneity, which is common in federated intrusion detection systems. For experimental consistency, both of the datasets are divided into a five-class classification problem, where one class represents benign traffic, and four represent aggregated attack categories

NSL-KDD and CICIDS2017 were selected because they represent different generations of network attacks. NSL-KDD contains legacy attack patterns while CICIDS2017 contains modern traffic-based attacks. This temporal diversity creates realistic cross-domain heterogeneity similar to real federated IDS deployments where organizations face different threat landscapes.

B. Data Preprocessing

Before federated training, both datasets undergo a preprocessing pipeline. Continuous features are normalized using Z-score normalization whereas label encoding is applied to the categorical attributes to convert them into numerical representations. Undefined, missing, and infinite values are replaced with numerical default values to maintain stability during training.

All feature vectors are projected into a fixed-dimensional input space of 79 features to maintain compatibility across datasets. The train–test split is performed in an 80:20 ratio to preserve class distribution. All clients follow the same preprocessing pipeline to mitigate preprocessing-induced bias..

C. Federated Client Simulation

The training data is partitioned into multiple client-specific subsets for federated learning. The experiments are conducted with 10 clients as a standard configuration. Both IID and Non-IID data distributions are considered to model realistic deployment scenarios. In the IID setting, samples are randomly selected and distributed uniformly across clients, resulting in statistically similar local datasets.

On the contrary, for the Non-IID setting, a Dirichlet-based partitioning strategy is used, introducing heterogeneity across clients. This setup shows real-world scenarios where organizations observe different attack frequencies and traffic profiles. For heterogeneous experiments, clients are differentiated by dataset origin and trained on both datasets. This design explicitly models cross-domain federated learning.

D. FedAvg Baseline Model

As a baseline, a federated averaging model is implemented using a neural network. It consists of a multi-layer feedforward network having residual connections, which help improve optimization stability. All model parameters are shared across clients, and the server aggregates them using standard federated averaging. This approach is commonly used in federated intrusion detection systems and serves as a reference point to evaluate the impact of personalization and privacy mechanisms.

E. Personalized Federated Learning Model

A personalized federated learning architecture is used to handle data heterogeneity. Each client maintains a local encoder that captures dataset-specific characteristics, while higher-level representations are aligned through a shared latent space. The dataset-specific local encoders transform raw feature vectors into intermediate latent representations, which pass through a shared alignment layer and a global classifier.

The local encoders are always kept private to each client. Only the shared alignment and classifier parameters are exchanged and aggregated during federated training. This strategy ensures that clients preserve domain-specific feature representations while benefiting from collaborative learning.

F. Federated Training Protocol

Federated training is conducted over 20 federated rounds in all experiments. Before every round begins the current shared parameters are broadcasted by the server to all the clients who are participating. Each client then performs local training for 2 epochs using mini-batch stochastic optimization with batch size 128. The baseline model uses Adam optimizer which has a fixed learning rate of 1×10^{-3}. Our model uses Adam optimizer with Cosine Annealing LR scheduling, which has initial learning rate as 1×10^{-3} and decays to minimum 1×10^{-5} over the training period.

Also, our method uses Focal Loss (γ=2.0) to address class imbalance and the baseline uses standard Cross Entropy loss. After local updates, the updated shared parameters are transmitted back to the server. The server aggregates them using federated averaging (FedAvg) algorithm and the updated shared model is sent back to the clients. Model performance is monitored using centralized evaluation at the end of each round.

G. Privacy-Preserving Mechanism

Differential privacy is incorporated into the federated training process to evaluate privacy–utility trade-offs. Before aggregation, shared model updates undergo gradient clipping followed by Gaussian noise injection. The magnitude of the injected noise is controlled by the privacy budget parameter ε. Privacy mechanisms are applied only to the shared parameters. This design ensures that sensitive local representations remain private Differential privacy experiments were conducted with $\varepsilon \in \{0.5, 1, 2, 5\}$, $\delta = 10^{-5}$, and clipping norm 1.0.

H. Evaluation Metrics

Model performance is evaluated using multiple metrics. Accuracy and macro-averaged F1-score are used to measure classification effectiveness. Operational reliability is measured using False Positive Rates (FPR). In addition, median inference latency is measured to evaluate deployment feasibility, and average communication cost per client is recorded to quantify federated overhead. These metrics determine the scalability, accuracy, robustness, and efficiency of the IDS framework.

I. Implementation Details

All experiments are implemented using the PyTorch deep learning framework. Federated learning is simulated in a centralized environment. For fair comparison, experimental configurations like privacy budgets and client partitions, are kept the same across all federated runs.

J. COMPARISON TABLE

Method	NSL-KDD Accuracy	CICIDS Accuracy	Worst Client Accuracy	Std. Deviation
FedAvg	70.75%	80.23%	18.41%	0.2390
FedAvg + DP	~45%	~60%	~12%	~0.28
Personalized FL	70.75%	80.23%	18.41%	0.2390
Proposed PP-PFL	**91.24%**	**80.30%**	**58.45%**	**0.1267**

VI. RESULTS AND DISCUSSION

A. Overall Performance Comparison

Table I presents the final performance comparison between the baseline personalized federated learning approach (CrossEntropy loss with fixed learning rate) and the proposed framework incorporating Focal Loss and Cosine Annealing LR scheduling.

Metric	Baseline	Proposed	Improvement
NSL-KDD Accuracy	70.75%	91.24%	+20.49%
CICIDS2017 Accuracy	80.23%	80.30%	+0.07%
Mean Client Accuracy	89.69%	90.27%	+0.58%
Worst-Case Client	18.41%	58.45%	+40.04%
Client Std Deviation	0.2390	0.1267	-46.99%
NSL-KDD F1-Score	41.43%	47.71%	+6.28%
CICIDS2017 F1-Score	17.81%	17.81%	+0.00%

TABLE I. PERFORMANCE COMPARISON: BASELINE VS PROPOSED FRAMEWORK

The results demonstrate substantial improvements across multiple dimensions. NSL-KDD accuracy increases by 20.49 percentage points, while the worst-performing client improves by 40.04 percentage points—allowing previously failing clients to participate effectively. The reduction in standard deviation from 0.2390 to 0.1267 indicates 47% improvement in fairness and stability across heterogeneous clients. The asymmetric improvements between NSL-KDD and CICIDS2017 reveal important insights: the proposed optimizations are most effective when addressing severe class imbalance and challenging non-IID distributions characteristic of NSL-KDD. CICIDS2017, being more balanced, does not require aggressive optimization, yet the proposed method maintains performance without degradation.

B. Ablation Study: Component Contributions

To evaluate the individual contributions of each optimization component, we conducted a systematic ablation study. Table II presents the results.

Configuration	NSL-KDD Accuracy	Worst-Case Client	Std. Deviation
Baseline (CrossEntropy)	70.75%	18.41%	0.2390
+ Focal Loss only	83.40%	10.93%	0.2621
+ Focal + Cosine LR (Proposed)	**91.24%**	**58.45%**	**0.1267**

TABLE II ABLATION STUDY: COMPONENT ANALYSIS

While Focal Loss alone improves NSL-KDD accuracy by 12.65 percentage points (70.75% → 83.40%), it paradoxically degrades worst-case client performance from 18.41% to 10.93% and increases variance to 0.2621. This occurs because Focal Loss emphasizes hard examples, causing optimization instability for severely imbalanced clients without proper learning rate control [9], [10].

When combined with Cosine Annealing LR, the framework achieves substantial improvements: NSL-KDD accuracy 91.24% (+20.49pp), worst-case client 58.45% (+40.04pp), and standard deviation 0.1267 (-47%). The scheduler prevents oscillations while maintaining learning capacity [11], [12].

Key Finding: Neither Focal Loss nor Cosine Annealing LR alone achieves robust performance under extreme non-IID conditions. Their synergistic combination is essential—Focal Loss provides class imbalance handling, while Cosine Annealing provides optimization stability.

C. Hyperparameter Sensitivity: The "Goldilocks Effect"

Table III shows sensitivity to personalization strength λ.

λ Value	NSL-KDD Accuracy	Worst-Case Client	Std Deviation
0.001	91.24%	19.95%	0.2322
0.01 (Optimal)	**91.24%**	**58.45%**	**0.1267**
0.05	91.24%	19.95%	0.2322

TABLE III. LAMBDA SENSITIVITY

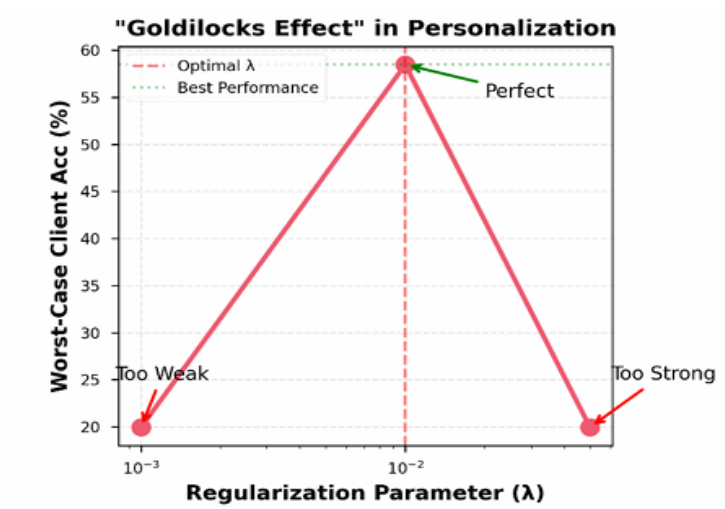

Fig. 2. Lambda Goldilocks curve

Figure 2 reveals a striking "Goldilocks effect": both weak (λ=0.001) and strong (λ=0.05) regularization produce identical poor worst-case performance (~20%), while optimal λ=0.01 achieves 58.45%.

D. Per-Client Performance Analysis

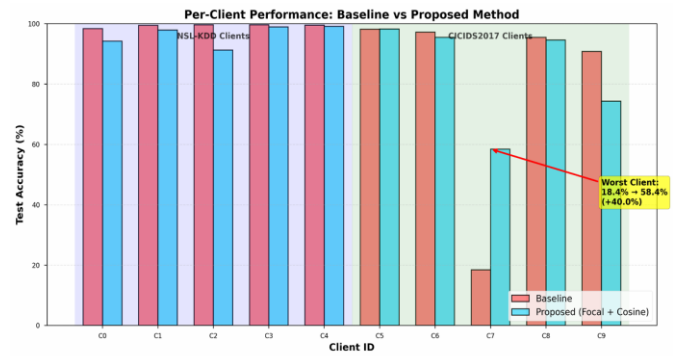

Fig. 3. Per-client bar chart

Figure 3 shows individual client performance. The proposed method dramatically improves NSL-KDD clients, particularly the worst-performing client: 18.4% → 58.4% (+40 points). CICIDS2017 clients maintain high performance (>90%), demonstrating improvements on challenging data don't degrade well-performing clients.

E. Convergence and Training Stability

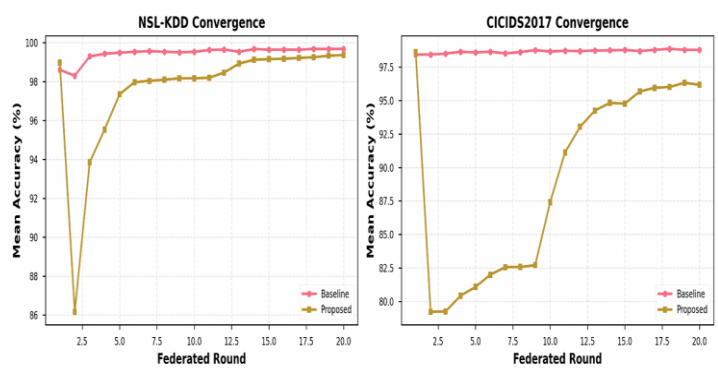

Fig. 4. Convergence comparison

Figure 4 shows NSL-KDD convergence. The proposed method achieves stable, consistent improvement reaching 91.24% final accuracy, while baseline plateaus at 70.75%. Similar patterns observed for CICIDS2017.

F. Privacy-Utility Tradeoff and Operational Metrics

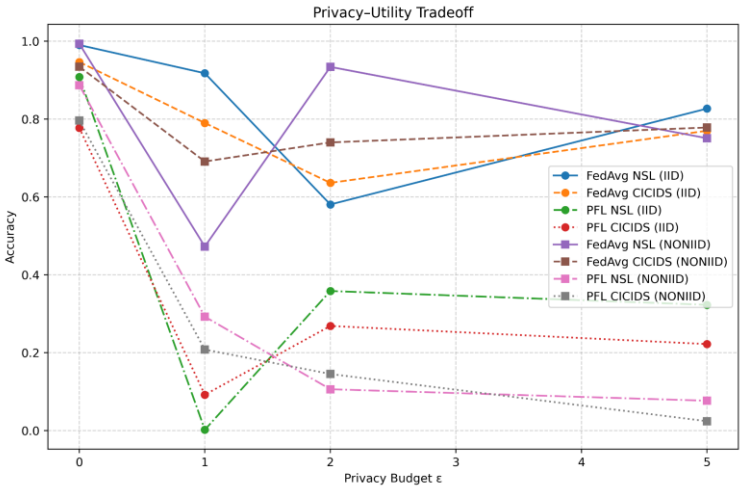

Fig. 5. Privacy–utility trade-off across different privacy budgets

Figure 5 shows accuracy vs privacy budget. Personalized FL exhibits smoother degradation under strict privacy constraints compared to FedAvg, maintaining consistent behavior without abrupt drops.

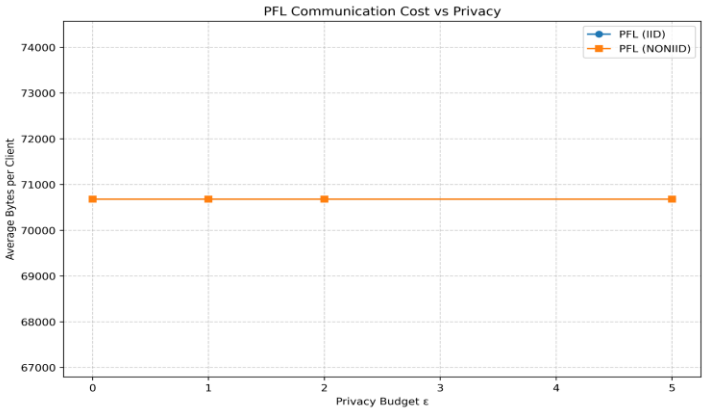

Fig. 6. Communication cost per client under different privacy budgets

Figure 6 shows communication cost remains constant across privacy budgets (~70KB/client), demonstrating differential privacy doesn't increase overhead. Median inference latency stays sub-millisecond (<0.01ms), confirming deployment feasibility.

G. Discussion and Key Observations

Three critical findings emerge. First, synergistic optimization is essential— Focal Loss alone harms fairness despite improving aggregate accuracy. Second, the "Goldilocks effect" reveals hyperparameter selection critically impacts fairness. Third, the framework enables equitable participation: 40-point worst-case improvement allows organizations with challenging data to participate effectively rather than being left behind. The asymmetric improvements (NSL-KDD +20%, CICIDS +0.07%) provide important insights: optimizations most benefit severely imbalanced datasets. For well-balanced data, standard approaches suffice, but the proposed method maintains performance without degradation.

VII. CONCLUSION AND FUTURE WORK

This paper addressed the challenge of fair and robust federated intrusion detection under heterogeneous data distributions. Standard FedAvg suffers severe degradation—worst-case client accuracy dropping to 18.41%—which compromises the fairness and purpose of collaboration. The proposed personalized federated learning framework, combining Focal Loss with Cosine Annealing LR, achieved substantial improvements, raising worst-case performance to 58.45%, improving NSL-KDD accuracy from 70.75% to 91.24%, and reducing variance by 47%. The observed "Goldilocks effect'' in personalization strength (λ) further highlights the importance of balanced regularization for equitable performance across clients. These results demonstrate that the framework enables resource-constrained organizations to benefit from federated IDS without being disadvantaged by data quality differences. Future work will focus on adaptive hyperparameter tuning, large-scale (100+ client) evaluations, adversarial robustness under Byzantine threats, and dynamic aggregation strategies to further strengthen fairness and practical deployment.

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